

RESEARCH ARTICLE

Power to the People! Assigning Popular Music Genres Based on Listeners' Perspective Using a Genre Tree Approach

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Abstract

Despite popularity of playlists on streaming platforms, music genres still remain a primary organizational tool among listeners, stakeholders, and researchers. Categorical genre assignments form the basis of many music genre recognition models in MIR and recommender systems research as well as in musicological basic research on music preferences and functions. Yet, 'correct' genre labels for popular music exemplars and structure of genre taxonomies are often contested, raising concerns about reliability, validity, and comparability of research results. In contrast, popular music studies have long emphasized the listener-dependent nature of genre assignments. In other words: Genre assignment may be considered a matter of majority voting within a listener community. So why not make a virtue of necessity? With the present contribution, we present a new methodology for constructing hierarchical genre trees from *MusicBrainz* folksonomy genre votes and demonstrate feasibility of this approach for 167,762 tracks listened to in Germany across the past 50 years. The resulting Community Music Genre Tree for popular music in Germany (COMGET-G) comprises 234 genres—interesting for cultural analyses, but probably too detailed for many applied research tasks. We address this by presenting an additional algorithm for folding such trees to a desired number of genre categories for any research problem (FOLDCAT). For automatic category assignment of music tracks still lacking community votes, we also develop ML classifiers at multiple granularity levels using openly-available music features and metadata from different online databases. Exemplary results indicate that our approach is able to assign a proper genre category representing a multiple listeners' perspective to any popular music track in a transparent, reproducible, and intersubjectively comprehensible way.

Keywords: music genre recognition, hierarchical taxonomy, folksonomy, network analysis, machine learning classification, German popular music

1. Introduction

Although playlist-driven listening has grown alongside the global diffusion of streaming platforms, genre labels still remain the primary way people and institutions organize popular music: Listeners search and browse by genre; playlist editors and DJs assemble genre-based playlists; A&R, marketing and music labels position and promote artists using genre tags; and music journalists and researchers rely on genre labels to describe and compare popular music (Bonini and Magaudo, 2024). Some scholars argue that classificatory practices have weakened since the late twentieth century (DiMaggio, 1987) and that certain genre

boundaries are unstable (Silver et al., 2016), but empirical evidence shows that listening and social signaling remain organized around genres (Airoidi et al., 2016; Lonsdale, 2021). On most music streaming platforms, genre assignments are embedded in metadata and pipeline logics—and hence used as search filters, playlist seeds, and features in recommender systems. Thus, platform algorithms amplify genres' practical significance for discovery and market positioning (Valverde, 2025). For researchers, conversely, genre labels serve both theoretical and methodological roles: they form variables in preference and reception studies (anonymized authors, XXXX), inputs to

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recommender-system research (Porcaro et al., 2024), and useful aggregator variables for measuring music diversity and mitigating hubness in high-dimensional audio and metadata feature spaces (Flexer et al., 2018; Flexer and Grill, 2016).

However, ‘correct’ genre labels for individual popular music phenomena (i.e., tracks, artists, releases, scenes, market, playlists, labels, performances etc.) are frequently contested (van Venrooij and Schmutz, 2018). Different actors may attach different genre labels to the very same music releases. Artists employ them to signify their musical identity; publishers and distributors tag releases for strategic market placement; critics and musicologists apply normative or historical taxonomies; and listeners use genre terms to organize their listening collections and to negotiate their distinctive musical identity in everyday communicative interactions (Frith, 1996; Negus, 1999). In result, not only do chosen genre labels diverge, but the taxonomical systems employed also differ strongly in degree of differentiation. For example, the commercial music metadata platform *Gracenote* uses a compact set of 25 genre tags (Jiang et al., 2024), whereas *Spotify* exposes many thousands of genre tags (McDonald, nd). Music databases and platforms also differ in terms of the objects to which they assign genres (track, release, or artist) and in assignment strategies (single primary genre versus multi-label tag sets).

These differences pose methodological challenges for empirical researchers across MIR and computational musicology. Problems encompass inconsistent genre labels that can degrade generalizability of supervised machine-learning models (Bogdanov et al., 2019), heterogeneously interpreted survey items in questionnaires on musical preferences (Lange et al., 2025), and, on a larger scale, the issue of noncomparability between studies - undermining replication and meta-analysis (Sturm and Flexer, 2023). In spite of these well-known challenges, researchers in MIR and computational musicology often pragmatically adopt available commercial or open taxonomies, without systematically engaging with the actual source of labels or underlying theoretical assumptions (Green et al., 2024). To address these problems, the present paper presents a theory-based, reproducible procedure to derive a scalable and transparently constructed genre taxonomy for any given corpus of popular music based on the community votes of a genre folksonomy. Paired with algorithmic music-genre recognition, this approach can (1) surface empirical regularities and disagreements in genre label use that may inform popular music studies, and (2) provide a shared, probabilistic ‘currency’ for cross-disciplinary work in computational musicology and MIR—so that algorithmic classifiers, preference studies, and sociological analyses may operate from a common, well-documented mapping.

1.1 Musical Genres as hierarchical discourse formation

While MIR and computational musicology often treat music genres as categories, in popular music studies they are in fact thought of as organized hierarchically. For example, Pachet and Cazaly (2000) identified genealogical links (e.g., *rock* as parent of *alternative rock*), geographical links (*hip hop* as supergenre of *German hip hop*), and historical-period links (*pop* as supergenre of *80s pop*) in existing taxonomies. As Brackett (2016, p. 8) argues, hierarchy depth is not limited to two levels but can extend virtually indefinitely. Hence, popular music genres are often visualized in trees (Crauwels, 2022).

However, scientific consensus on which labels to select and how to relate them is seldom achieved. Even expert-constructed taxonomies show limited agreement (Pachet and Cazaly, 2000), and expert classifications often diverge from listener community norms (Sordo, 2008). Adopting the power-critical perspective of cultural studies (du Gay et al., 2013), we interpret the discourse on genres in the popular music field through Foucault’s (2013) concept of *discourse formation*. From this viewpoint, artists, musicologists, industry, and listeners exert unequal influence over which genre labels circulate and stabilize in the long run (Lena and Peterson, 2008). Artists often self-assign genres as strategic acts of boundary-making to gain recognition and position themselves in scenes and markets (Negus, 1999). Genres assignments by musicologists might partly mirror normative decisions rooted in struggles over cultural authority while also serving pedagogical and archival purposes (Bourdieu, 1985).

Industry actors’ genre assignments mainly reflect how the music industry strategically positions and markets artists and their releases within the field which may often-times diverge from listener perception (Valverde, 2025). We term these industry-driven assignments *distribution genres*. By contrast, aggregating listener votes treats genre assignment as implicit majority voting within a community, privileging the perspective of those who engage with the music rather than those who distribute or normativize it. This democratic aggregation replaces single-source genre assignments from label A&R departments, artists, platform editors, or individual researchers with collective consensus, capturing how listeners actually categorize music in everyday discourse. We term these listener-derived assignments *community music genres*, reflecting the emergent consensus of listening communities rather than the view of economically powerful institutional gatekeepers and high-authority cultural actors.

To empirically ground genre assignments in listener discourse rather than in industry, producer or musicological authority, we propose a new methodological approach that derives a *community music genre* hierarchy and *community genre* assignments for music tracks from the votings of global music lis-

tener communities, also termed *folksonomies*. Platforms such as *RateYourMusic* (nd), *Discogs* (2024), or *MusicBrainz* (nd) provide crowdsourced ‘ground truth’ on the genre of musical objects reflected in aggregated listener tagging behavior. *MusicBrainz* seems particularly suitable for our endeavor: it maintains a detailed whitelist of over 2,000 genre tags without imposing an a-priori hierarchy, and its democratic voting system allows users to assign and vote on tags at multiple levels (track, release, artist), enabling nuanced analysis of consensus and disagreement.

1.2 Genre taxonomy requirements for MIR and computational musicology

For the application of genre taxonomies in MIR research and computational musicology, several research-pragmatic requirements have to be taken into account. First, balanced genre distributions improve statistical power and machine learning performance (Japkowicz and Stephen, 2002). However, subgenres typically receive fewer track assignments than their super categories (Bogdanov et al., 2019), creating an inherent imbalance propagating through hierarchical levels. A novel method to empirically derive genre taxonomies from hierarchical ground-truth should therefore optimize balance within each hierarchy level separately, constructing well-balanced tree-hierarchies where sibling genres exhibit comparable sizes regardless of their position in the hierarchy.

Second, different research questions may require different numbers of *genre categories* or, as we term it, a different degree of *genre granularity*. Granularity refers to the level of detail at which genres are differentiated: low granularity collapses many subgenres into few broad categories (e.g., *rock*, *pop*, *electronic*), whereas high granularity preserves fine-grained distinctions (e.g., *death metal*, *synthwave*, *UK garage*). Surveys of musical taste may require moderate to high granularity (Lange et al., 2025, e.g., 24 genres), machine learning models for genre classification often perform better with fewer, broader categories (Genussov and Cohen, 2010), and subcultural analyses may demand very fine-grained subgenre differentiation (Caparrini et al., 2020). Since preserving consistency with the overall hierarchical ground truth of genre assignments remains desirable, this implies a method to derive genre categories at multiple scales from a single underlying tree-hierarchy.

Third, derived genre category assignments of whatever granularity should demonstrate sufficient validity, consistency and generalizability to popular music artist, tracks and releases beyond the ground-truth employed for constructing the category system (Sturm and Flexer, 2023). External validation may be realized by comparing genre assignments with those commonly used in MIR and computational musicology. Other aspects of validity, consistency of assignment and out-of-

sample generalization may be demonstrated by training automated classifiers for the newly-created genre category system on different granularity levels drawing on openly available music features and metadata.

1.3 Feature space for community music genre recognition

Algorithmic classification of musical genres, or music genre recognition, is considered a core task in Music Information Retrieval. Green et al. (2024) provide a systematic state-of-the-art review and also critique how genre theory from the social sciences and humanities is unfortunately often overlooked when selecting features (for a systematic review on genre theories see Merlini, 2020). Problematically, researchers in MIR and music psychology often-times assume that genre can be inferred from audio features alone (Tzanetakis and Cook, 2002; Rentfrow et al., 2011). However, this conflicts with a sociological approach to genres, which we also adopt here, describing them as “systems of orientations, expectations and conventions that circulate between industry, text and subject” (Neale, 1980, p. 19), emphasizing extra-musical context alongside sound as determining factors for genre assignments. Franco Fabbri’s (1982) genre theory offers such a synthesis: musical style (instrumentation, rhythm, harmony, timbre) and release milieu (subcultural context, artist positioning, market placement) together inform how listeners assign genres. While MIR toolboxes extract style features from audio signals (Bogdanov et al., 2013), open music databases such as MusicBrainz (MusicBrainz, nd), Genius (Genius, 2009), etc. provide distributional metadata (label, chart performance, platform placement, lyrics). For our theoretical endeavour that focusses on genre assignments by listeners, drawing on such metadata appears indispensable.

Song lyrics occupy an intermediate position, conveying both, emotional expression and stylistic content (Brand et al., 2019), as well as contextual information on release milieu through employed language codes and topics that signify artists’ subcultural affiliation (Carbone et al., 2024). Further, a genre classifier might need to pay special attention to the *mainstream pop* or *pop* metagenre that Fabbri (1982) describes as encompassing eclectic, commercially optimized works that resist association with a specific milieu. Artists from former niche scenes are often retrospectively relabeled as *pop* or *mainstream* after achieving commercial success (Holt, 2007, p. 20). Therefore, popularity itself appears to be a relevant feature dimension for classification. In sum, the anonymized authors (XXXX) recently proposed a feature space for algorithmic music genre classification spanning four dimensions—musical style, release context, lyrical content, and popularity to operationalize the multi-dimensional constitution of genre following Fabbri’s (1982) ap-

proach.

1.4 Research Questions

Drawing on the theoretical and methodological foundations outlined above, the present work pursued three research questions:

RQ1 Can we construct an empirically grounded and sufficiently balanced hierarchical genre tree from multi-tag folksonomy votes for a given popular music corpus?

RQ2 Can we aggregate the inferred tree structure to a scalable number of distinct genre categories suitable for machine learning and statistical applications?

RQ3 How valid and generalizable are community-derived genre category assignments for empirical music research?

a) To what extent do the community-derived categories agree with existing taxonomies of comparable granularity (e.g., Rosamerica, Discogs, Deezer)?

b) Can algorithmic ML-classifiers reliably predict community-derived genre categories at multiple granularities employing openly available music style, release-milieu, lyrical, and popularity features?

In the following sections, we will hence introduce a method to create a balanced hierarchical genre tree based on genre co-occurrences in folksonomy tags (COMGET), describe a tree folding algorithm to flexibly create genre category systems with different levels of granularity for various research tasks (FOLD-CAT), present comparisons of genre assignments with existing taxonomies and results of algorithmic classifier training for different granularities, and finally discuss our findings and summarize results.

2. RQ1: Creating a Balanced Hierarchical Genre Tree

This section describes method and exemplary results of the novel COMGET procedure that creates a balanced hierarchical genre tree for any popular music dataset based on genre co-occurrences in folksonomy tags.

2.1 Exemplary Data: Popular Tracks in Germany dataset (POPTRAG) and MusicBrainz Community genre tags

As initial demonstration of our new methodological approach, we perform the analysis depicted in this paper with a corpus of 254,334 unique popular tracks released and frequently listened to in Germany over the past 50 years (POPTRAG). Four sources comprise this corpus: 215,702 tracks from weekly *MediaControl* German single and album charts (18 Nov 1977–21 Nov 2024; reduced to 75% where linkage success to *Spotify* identifiers was possible); 26,949 *Spotify* track recommendations from 126 German-market genre seeds (retrieved 29 Nov 2024; median 325 per seed, IQR 25, range 49–343); 8,801 tracks from 100 top-featured

Spotify playlists for the German market (retrieved on 29 Nov 2024, median 90 tracks per playlist, IQR 42, range 28–750); and, finally, 11,994 unique track entries from *Germany's Daily Top 200 Spotify* charts (01 Jan 2017–30 Nov 2024). From all sources, we excluded tracks shorter than 30 seconds (pertains to 853 tracks, likely non-musical snippets) and retained only unique title-artist combinations, preferring earlier release dates in case of duplicates.

In a second step, we acquired *MusicBrainz* track genre tags for all music tracks in the POPTRAG corpus. Hereto, only genre labels contained in the *MusicBrainz* whitelist (snapshot 16 Jun 2025) were taken into account, reduced by tags referring to genres beyond popular music (*non-music*, *interview*, *comedy*, *nature sounds*, *children's music*, *spoken word*, *standup comedy*). In case track-level tags were missing, we employed release tags instead. If these were missing, too, we used the artist genre tags. For denoising, we then required any genre tag to appear with at least 50 distinct artists in the overall corpus to ensure sufficient discursive support for any genre, if not, they were discarded. Finally, we retained only tracks in the corpus exhibiting at least two community genre tags fulfilling all of these criteria, leading to a removal of 86,613 tracks (34%) from the corpus and yielding a *MusicBrainz*-tagged subset of $N = 167,721$ tracks annotated with 234 distinct music genre tags spanning release years 1924–2024 (median 2012, IQR 16 years). Community genre assignments comprised between 1 and 22 different tags per track (median 4, IQR 3) and cast between 2 and 135 votes per track (median 6, IQR 7).

2.2 Methods: Hierarchical Genre Tree Construction (COMGET)

Based on the genre-tagged and denoised POPTRAG corpus, we algorithmically built a hierarchical genre tree from *MusicBrainz* tag co-occurrences, adapting and further developing a method originally presented by Schreiber (2015; 2016). Thereto, we first constructed a weighted co-occurrence network where nodes represent *genre* labels and directed edges $e_{A \rightarrow B}$ quantify how strongly *genre A* predicts *genre B* on the same track. For any ordered pair (A, B) , we computed a vote-weighted conditional frequency:

$$e_{A \rightarrow B} = P(B | A) \times \frac{1}{|T_{A,B}|} \sum_{t \in T_{A,B}} \frac{v_{t,B}}{v_{t,\text{total}}} \quad (1)$$

where $T_{A,B}$ denotes tracks tagged A and B , $v_{t,B}$ the votes for B on track t , and $v_{t,\text{total}}$ the total votes on t . This formula produces asymmetric edges ($e_{A \rightarrow B} \neq e_{B \rightarrow A}$) that combine directional co-occurrence with community vote strength. To illustrate: suppose 1,000 tracks carried *metal* and 600 also carried *rock*, giving $P(\text{rock}|\text{metal}) = 0.6$. If the average vote share for *rock*

on co-tagged tracks equalled 0.5, then $e_{metal \rightarrow rock} = 0.6 \times 0.5 = 0.3$. Conversely, if 6,000 tracks carried *rock*, 600 of those carried *metal*, then $P(metal|rock) = 0.1$; with an average *metal* vote share of 0.7 on *rock* tracks, $e_{rock \rightarrow metal} = 0.07$. Hence, calculated directed probabilities express both prevalence and tag confidence at the same time.

In a second step, we resolved reciprocal track connections to obtain a unidirectional scaffold. For each pair of tags, we thereto subtracted the weaker directional probability from the stronger and kept only the positive residual:

$$e'_{A \rightarrow B} = (e_{A \rightarrow B} - e_{B \rightarrow A})_+ \quad (2)$$

$$e'_{B \rightarrow A} = (e_{B \rightarrow A} - e_{A \rightarrow B})_+ \quad (3)$$

In the example this yields $e'_{metal \rightarrow rock} = 0.6 - 0.07 = 0.53$ and $e'_{rock \rightarrow metal} = 0$, making *metal* $\rightarrow$ *rock* a clear candidate supergenre link.

Third, we enforced a strict tree-like structure by assigning to each node its single strongest outgoing edge as its *supergenre*; while all other edges were discarded. Components without *supergenre* were attached to an artificial root node *POPULAR MUSIC* to guarantee connectivity of any genre tag to the overall tree. In the example, *metal* would attach to *rock* unless a stronger edge existed, and *rock* would attach to the root if it had no *supergenre*.

Finally, we balanced the tree in terms of optimizing leaf-node populations on each hierarchical level. Thereto, we estimated each genre node's size as the sum of its assignment probabilities across tracks in the corpus, then computed the Gini coefficient within each hierarchical tree-level for all genre nodes contained, and defined overall tree imbalance as the weighted mean of all tree-level Ginis. We then searched for the least-imbalanced tree by iteratively removing the weakest hierarchical link of the whole tree, reattaching the resulting dangling tree to *POPULAR MUSIC*, and recomputing imbalance again; we then finally selected the iteration with minimal overall tree imbalance.

2.3 Results: Community Genre Tree for Popular Music in Germany (COMGET-G)

The initial POPTRAG *genre* co-occurrence network contained 234 *genre* nodes forming a single connected component with 45.14% density, indicating relatively high inter-tag connectivity. By median, each *genre* co-occurred with 97 different others (IQR 63.5; range 3 for *waltz* to 229 for *pop*), revealing large differences between niche and ubiquitous genre tags. After resolving reciprocal directions, each *genre* had on average 29.5 incoming subgenre links (IQR 32; min 0; max 228 for *pop rock*) and 55 outgoing supergenre links (IQR 32; min 0 for *rock*; max 105 for *dub*).

The balance optimization procedure (Figure 1) reattached *pop* and *hip hop* directly to *POPULAR MU-*

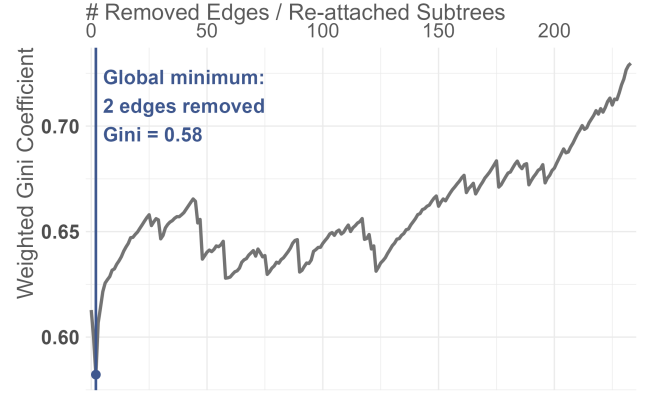


Figure 1: Weighted Gini coefficient (0–1) across iterations of cutting weak genre links and reconnecting dangling subtrees to *POPULAR MUSIC*.

Level	# nodes	# tracks	Gini
1	3	total: 47,846 (29%) 12,472; 16,401; and 18,973	0.136
2	119	total: 90,381 (54%) median: 263 IQR: 789 range: 10.4 to 5832	0.643
3	110	total: 29,343 (17%) median: 156 IQR: 259 range: 14.7 to 1496	0.531
4	2	total: 192 (0.1%) 55.6 and 136	0.420

Table 1: Structure of COMGET-G at different hierarchy levels; track counts are sums of genre tag assignment probabilities in POPTRAG.

SIC—*pop* had formerly been subordinate to *rock*, and *hip hop* to *pop*, however these connections were the weakest in the whole tree and their cutting resulted in clear improval of overall balance and also appeared to make sense from the perspective of popular music taxonomies, where these genres are typically rather considered supergenres. The three weakest remaining links are *electronic* to *pop* (0.22), *classical* to *pop* (0.27), and *jazz* to *pop* (0.29) - however these three were not populated enough with corpus tracks to form independent supergenres directly connected to the root. The resulting *Community Music Genre Tree for popular music in Germany* (COMGET-G, provided for interactive exploration in the digital appendix) is highly leaf-heavy: 89.36% of genre-nodes lack subgenres. *Rock* emerges as the largest *supergenre* with 63 direct subgenres, and only *rock*, *pop*, and *hip hop* remain connected directly to *POPULAR MUSIC*, forming three broad parent genres. The tree exhibits four hierarchy levels (excluding the artificial root); see Table 1 for structural details.

3. RQ2: Creating Genre Category Systems with Scalable Granularity

This section describes procedure and exemplary results of the novel FOLDCAT tree folding algorithm that flexibly turns COMGET-type hierarchical genre trees into genre category systems with desired level of granularity.

3.1 Methods: Folding Algorithm (FOLDCAT)

To turn any hierarchical genre tree into a scalable number of broader *genre categories* suitable for given research problems and machine learning applications, we developed a greedy folding algorithm (FOLDCAT) with three objectives: optimal *genre category* balance, a user-selectable number of *genre categories*, and sufficient track counts per *genre* for robust machine learning and statistical analysis. To realize these, the FOLDCAT algorithm iteratively processes leaf *genres* beginning with those deepest in the hierarchy and largest in track count, continually moving upwards to the root. For each leaf *genre*, it checks whether a minimum track count threshold is met. This is computed as the sum of vote-weighted assignment probabilities across all tracks corresponding to the leaf. If it does and its supergenre can sustain removal, the algorithm keeps that leaf *genre* as a *genre category*. Otherwise, the algorithm absorbs the leaf *genre*—along with its siblings and any descendants already processed—into its supergenre by remapping all track assignments upward. This folding procedure continues until all remaining *genres* meet the minimum threshold. For a desired number of *genre categories*, users may then select the folding solution corresponding to a local Gini minimum, drawing on a hierarchy-level-weighted Gini coefficient (see section 2.2), thereby ensuring well-

balanced *categories* across all hierarchy levels.

For final category assignment, each track in the corpus obtains a single *genre label* via majority voting. Thereto, all original genre tags are mapped to their *category assignment* in folded COMGET, then each track receives the *category* tag with the highest summed votes. Ties are resolved by selecting the *genre category* with lower total assignment probability in the corpus thereby favoring more detailed *genre category* assignments. A detailed flowchart of the FOLDCAT algorithm is provided in the appendix. For the present exemplary analysis using the POPTRAG corpus turned into COMGET-G, we conducted a grid search across minimum *genre sizes* from 1,000 to 15,000 in increments of 5 and then selected four different local Gini minima from the results for comparison.

3.2 Results: Germany's Popular Music Genre Categories

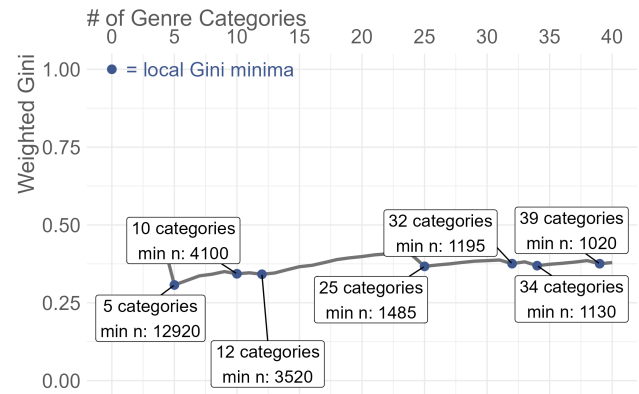


Figure 2: Weighted Gini coefficient (0–1) across COMGET-G folding solutions for minimum track counts from 1,000 to 15,000 in increments of 5, using the POPTRAG corpus.

For our POPTRAG corpus, the folding algorithm identified multiple locally-optimal categorical solutions for COMGET-G (Figure 2): a low-granularity solution with 5 *genre categories* (min. 12,930 tracks, Gini 0.307), medium-granularity solutions with 10 and 12 *genre categories* (min. 4,100 and 3,520 tracks; Gini 0.343 and 0.342), a high-granularity solution with 25 *genre categories* (min. 1,485 tracks, Gini 0.367), and solutions with higher granularity, comprising 32, 34, and 38 *genre categories* (min. 1,195, 1,130, 1,020 tracks; Gini 0.376, 0.369, 0.375). We focussed subsequent analyses on solutions with 5, 12, 25, and 32 *genre categories*, representing different granularities suitable for different types of research problems (Figure 3).

COMGET-G5 comprises *rock*, *pop*, *hip hop*, *electronic*, and *metal* (in descending order of track count). In COMGET-G12, *jazz* and *schlager* emerge as further distinct *pop subgenre categories*, while *rock subgenre categories* including *heavy metal*, *alternative rock*, *pop*

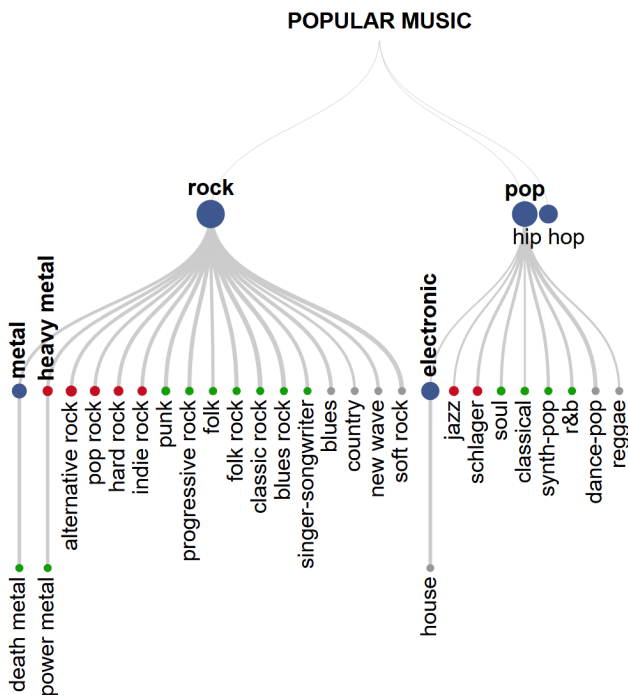


Figure 3: Genre category systems for POPTRAG corpus at four different granularities (5, 12, 25, 32 *genre categories*), projected onto the original COMGET-G tree hierarchy. Colors indicate additional categories introduced at each finer levels of granularity (blue = 5; red = 12; green = 25; grey = 32). Node size represents approximate track count in POPTRAG for the most granular solution with 32 categories (sum of vote-weighted assignment probabilities).

rock, hard rock, and indie rock appear at the same level. Notably, the majority of MusicBrainz community members did not classify *heavy metal* tracks as a *subgenre category* of *metal* but rather as a direct *subgenre category* of *rock* on the same hierarchy level as *metal*.

COMGET-G25 introduces further *rock subgenre categories* (punk, progressive rock, folk, folk rock, classic rock, blues rock, singer-songwriter) and additional *pop subgenre categories* (soul, classical, synth-pop, r&b). A third hierarchy level appears with *death metal* as a *subgenre category* of *metal* and *power metal* as a *subgenre category* of *heavy metal*. COMGET-G32 adds *house* as an *electronic subgenre category* and additional subgenre categories in the *rock* and *pop* branches (blues, country, new wave, soft rock, dance-pop, reggae). *hip hop subgenre categories* appear only in more fine-grained solutions that we do not document here (*pop rap* as 33rd genre, *trap* as 38th genre).

4. RQ3: Demonstrating Validity, Reliability and Generalizability of Results

Drawing on the the POPTRAG corpus of German popular music and MusicBrainz community genre tags, this section presents empirical comparisons of track genre assignments resulting from our novel community music genre categorization approach (COMGET + FOLD-CAT) with those resulting from genre category systems frequently used in MIR and computational musicology. Furthermore, it provides results of algorithmic genre classifier training we performed with results of the same novel community genre category systems. Both analyses were conducted with different genre granularities.

4.1 Methods: Validation Strategy, Predictor Variables, Machine Learning Workflow

For both analyses described in this section we derived categorical genre assignments for each track in the denoised and MusicBrainz-tagged POPTRAG corpus at four different levels of granularity (COMGET-G5, COMGET-G12, COMGET-G25, COMGET-G32) using COMGET and FOLDCAT as described in the previous section.

For a discussion of content validity of our community genre taxonomy solutions, we then employed contingency tables to compare results with genre classifications of the same tracks in existing and frequently used taxonomies with similar granularity. For this means, we retrieved *Rosamerica* genre classifications from *AcousticBrainz* (Alonso-Jiménez et al., 2020), community genre assignments from *Discogs* (2024) folksonomy, and genres assignments of the commercial music streaming platform *Deezer* (nd) for each track in our corpus, drawing on our own music data linkage tool *Spotilink* (XXXX). For also comparing genre balances, we additionally calculated category prevalences and overall gini coefficients for each taxonomy. For

easier interpretation of resulting charts and figures, we removed genre categories with less than 2.5 percent tracks in the corpus for these analyses.

To determine how well an algorithmic classifier would predict the COMGET-G *genre categories* identified by FOLDCAT, we trained independent classifiers for each of the four solutions. To that end, we again employed *Spotilink* to augmented POPTRAG with 192 descriptor variables covering *musical style*, *release context*, *lyrical content*, and *popularity* retrieved from the music platforms *Spotify*, *MusicBrainz*, *AcousticBrainz*, *Genius*, and *Deezer* by using ISRC matching or fuzzy string matching and employing either API-access or web scraping. Specifically, we retrieved *AcousticBrainz* high-level *audio features*; *artist demographics* (*persona*, *origin*, *birth year*, *gender*) from *MusicBrainz*; *popularity* metrics from *Spotify* and *Deezer* and finally *Echo-Nest* high-level *audio features* from *Spotify*. For lyrics, we constructed XXXX bag of words, categories of topic modelling XXXX. The full list of features is provided in table XX in the appendix. For testing to which degree music industry genre marketing influences how listeners ultimately categorize music we also included distribution genres into the predictor features. Thereto, we applied COMGET and FOLDCAT methodology on the 69 most prevalent *artist-level genre* tags from *Spotify* in our corpus and used resulting industry genre assignment as additional predictor variable in our model.

In our machine learning workflow, we first discarded tracks with more than 40% missing values on any variable, retaining 155,093 tracks (92%). We then performed an 80/20 train–holdout split of all observations on artist level to mitigate possible artist leakage effects (Flexer and Schnitzer, 2010). For training our classifiers, we employed 5-fold cross-validation (also group-split on artist level) for hyperparameter tuning. To reduce sparsity and high-cardinality effects in categorical predictors, we drew a 20% subsample of the training set, created artist-grouped CV folds on that subsample, and relabeled any factor level that occurred fewer than 100 times in at least one fold as “other”. Predictors that collapsed to a single “other”-level as well as two-level predictors where the “other” level had fewer than 100 observations were removed. This preprocessing reduced the number of variables from 192 to 149. All remaining categorical predictors were dummy-coded, using the most common level as the reference.

Further preprocessing was applied in sequence within each single fold: median imputation, adaptive class balancing (excluded for holdout set), and normalization. Adaptive class balancing employed a single tunable *target-ratio* parameter governing rebalancing degree. The algorithm computed a reference count from outcome class frequencies (median for ≤ 10 classes; 60th percentile for 11–20 classes; 65th percentile for > 20 classes), then downsampled classes ex-

ceeding $\text{reference} \times \text{target ratio}$ using NearMiss (Zhang and Mani, 2003) and upsampled classes below $\text{reference} \div \text{target ratio}$ using Borderline SMOTE (Han et al., 2005) or regular SMOTE when sufficient danger observations were unavailable. This approach preserved more data than fixed-ratio sampling while remaining tunable: *target ratio* = 1 enforced perfect balance; higher values retained original imbalance. The reference-count heuristic scaled to problem complexity, preventing excessive downsampling at higher resolutions.

For hyperparameter search in terms of this target-ratio and learner algorithm settings, we employed Bayesian optimization (Snoek et al., 2012) with a Latin hypercube initial grid (Husslage et al., 2011), early stopping if no improvement occurred after 10 iterations, and exploration of uncertain hyperparameter regions after 5 iterations without improvement. The final hyperparameters were selected over all tunes using the one-standard-error rule (Hastie et al., 2009). We tuned hyperparameters across five folds using the macro-averaged F1 score (mean per-class F1) to treat all *genres* equally during learner evaluation.

During initial hyperparameter tuning, we evaluated four different supervised ML algorithms based on a 20% subsample of the training data at low (5) and medium (25) granularity to identify the most suitable learner out of 4 options: elastic net (GLMNET; Tay et al., 2023), regularized discriminant analysis (RDA; Weihs et al., 2005), random forest (RF; Wright and Ziegler, 2017), and light gradient boosted machine (LightGBM; Shi et al., 2025). For each we tuned key hyperparameters: GLMNET — *penalty* and *mixture* in $[0, 1]$; RDA — *covariance fraction* and *identity fraction* in $[0, 1]$; RF — *feature fraction considered at each split* in $[1, \# \text{ of predictors}]$ and *leaf node size* in $[1, 100]$; number of trees was set to 500 for tuning and 1000 for final models; LightGBM — *boosting rounds* in $[50, 2000]$, *learning rate* in $[0.001, 0.1]$, *leaf node size* in $[1, 100]$, *tree depth* in $[3, 15]$, and *feature fraction* in $[1, \# \text{ of predictors}]$. For all learners, the target ratio used for adaptive balancing was tuned in between $[1, \max(\text{observations per class})/\min(\text{observations per class})]$. To identify the best learner, we employed a coarse hyperparameter search with a 10-level initial grid and a maximum of 25 iterations for Bayesian optimization.

After selecting the learner with best performance across folds, we performed a more extensive hyperparameter search with a 20-level initial grid and a maximum of 50 iterations with 5-fold crossvalidation on the full training set for all four levels of genre granularity (5, 12, 25, 32), trained the final classifiers on the full training set, and evaluated results on the holdout set. Then we calculated model-based feature importances and plotted confusion matrices for interpreting classification outcomes. Here, *Community certainty*

(overall and for single categories) served as a reference point for evaluating final classifier performance, since ML models logically cannot exceed the consistency of the underlying human community assignments (Sturm and Flexer, 2023). For this comparison *certainty* was computed as the mean assignment probability across all MusicBrains votes for each track that had been mapped to the *single-label genre* during the FOLDCAT procedure that built the category systems. All random operations in the machine learning workflow used the seed 42 for reproducibility.

4.2 Results 1: Comparison of COMGET-G variants with existing genre taxonomies

The *Rosamerica* genre classification system (XXXX) comprises eight categories, making it most comparable to COMGET-G5; however, *AcousticBrainz* lacked entries for 28% of POPTRAG tracks, introducing a modest popularity bias into this analysis. The resulting contingency table (Figure 4a) shows the strongest alignment between COMGET *metal* and *Rosamerica Rock* (83%). Other COMGET-G5 categories are distributed across multiple *Rosamerica* genres: COMGET *rock* mostly maps to *Rock* (45%), *Pop* (21%), and *Rhythm and Blues* (17%); *pop* maps primarily to *Pop* (35%) and *Rhythm and Blues* (29%); *electronic* maps to *Dance* (28%), *Rhythm and Blues* (21%), and *Pop* (19%), with a smaller share in *Hiphop* (14%); and *hip hop* maps to *Hiphop* (50%) and *Rhythm and Blues* (29%). Except for *metal*, all COMGET-G5 categories overlap with *Rhythm and Blues*, suggesting shared sound attributes in some tracks within these categories and a distinctive sound of *metal*. Given the broad scope of COMGET-G5 and the subgenres composing its categories, these mappings appear reasonable. The category balance of *Rosamerica* is marginally better than COMGET-G5's ($Gini_{COMGET-G5} = 0.488$, $Gini_{Rosamerica} = 0.417$) for the POPTRAG corpus.

Discogs' 15-genre taxonomy approximates COMGET-G12 in scale. For approximately 5% of tracks with COMGET-G assignments, we found no *Discogs* counterpart, introducing minimal popularity bias. The contingency table (Figure 4b) reveals strong alignment (> 80%) for all COMGET-G12 *categories* with a single *Discogs genre*, except for *pop rock* (max. 77%), *schlager* (max. 76%), *jazz* (max. 71%), and *pop* (max. 30%). Even where *Discogs* includes a *Jazz* category, COMGET-G's *jazz* overlaps substantially with *Discogs' Electronic* (11%) and *Pop* (8%). Similarly, *pop rock* is distributed across *Discogs' Rock* (77%), *Pop* (12%), and *Electronic* (8%); *schlager* emerges in *Pop* (76%, similar label to its COMGET-G *supergenre*) with an overlap to *Rock* (16%). Five *Discogs* categories—*Blues*, *Brass & Military*, *Children's*, *Non-music*, and *Stage & Screen*—show minimal correspondence to any COMGET-G12 *genre category*, mostly consistent with our noise-reduction filtering and focus on pop-

ular music. COMGET-G's *pop* category is distributed widely across *Discogs genres* (*Electronic* 30%, *Pop* 20%, *Rock* 22%, *Funk / Soul* 10%), reflecting pop's eclectic character. Seven of twelve COMGET-G12 *genre categories* overlap most strongly with *Discogs' Rock*, suggesting *Discogs* lacks granular rock sub-genre category labels needed to represent German popular music adequately. At this medium granularity, COMGET-G12 *category* assignments show much a much more balanced distribution for POPTRAG than *Discogs' assignments* ($Gini_{COMGET-G12} = 0.540$, $Gini_{Discogs} = 0.749$).

Employing the *Deezer* genre taxonomy we found no assignment for 9% of POPTRAG tracks, while 26 *Deezer* genres appear in the corpus; the contingency table (appendix Figure B.1) shows > 80% alignment only for *blues rock* → *Rock* and *hip hop* → *Rap/Hip Hop*. Most COMGET categories distribute across two or three *Deezer* labels (10% threshold). In contrast, *electronic* maps to *Electro* (34%), *Pop* (24%), *Dance* (20%), and *Alternative* (12%). Also, *folk* maps to *Pop* (29%), *Alternative* (21%), *Rock* (13%) and *Folk* (12%). In particular, *pop* aligns strongly with *Deezer's Pop* category (68%). COMGET-G25 categories are substantially more balanced than *Deezer* ($Gini_{COMGET-G25} = 0.602$, $Gini_{Deezer} = 0.753$).

4.3 Results 1: Classification Performance for four different COMGET granularities

Learner comparison (Figure 5) shows that all four supervised machine learning algorithms tested perform within a similar range on the 20% subsample (mean $F1_{macro}$; CV = 0.717 – 0.758 for COMGET-G5 and $F1_{macro}$; CV = 0.468 – 0.503 for COMGET-G12). However, LightGBM distinctively emerged as the strongest performer at both granularities (COMGET-G5 and COMGET-G12) and was thus selected as the final learner for all subsequent analyses. While GLMNET ranked second best for COMGET-5, both GLMNET and RDA performed worse for COMGET-G12 compared to tree-based methods, suggesting that models limited to primarily linear or low-order quadratic decision boundaries struggled to capture the increased complexity at finer outcome granularity.

For the POPTRAG dataset classified with COMGET-G5, LightGBM models (Table 2) achieved $F1_{macro} = 0.764$ (SE: 0.004) across CV folds and 0.743 on the hold-out set, indicating slight overfitting. Per-class F1 correlates strongly with *community assignment certainty* ($Cor_{Kendall} = 0.949$). Given mean *community certainty* of 90%, the classifier performs satisfactorily. Cohen's $\kappa = 0.674$ on the hold-out set ($\kappa_{CV} = 0.695 \pm 0.007$) is very close to established inter-rater agreement standards ($\kappa = 0.7$) in the social sciences (Hildebrandt et al., 2015, p. 327). The confusion matrix (Figure 6a) reveals the classifier struggles most with genre *electronic* (56% true positive-TP), often confusing it with

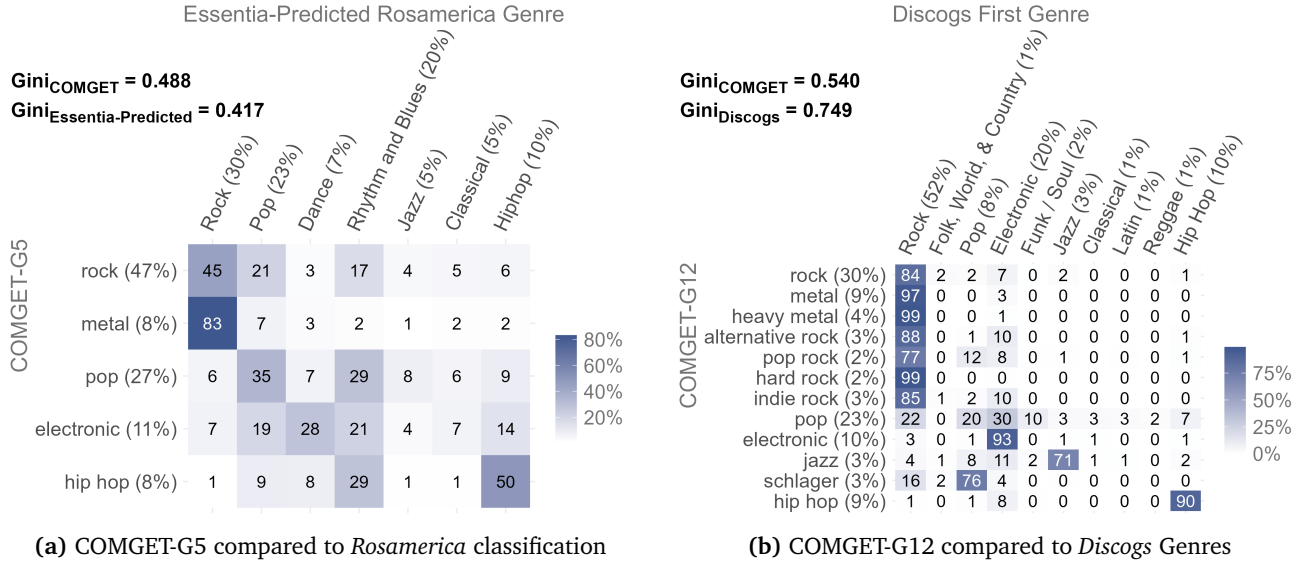


Figure 4: Contingency tables comparing COMGET-G genre assignments folded with granularity 5 or 12 to corresponding genre classification by taxonomies commonly used in MIR and computational musicology; values show $P(\text{External} | \text{COMGET})$. Additionally, we calculated relative genre prevalences and Gini coefficient across genres. Categories with less than 2.5% relative frequency in the corpus were discarded (*Rosamerica*: *Speech*, *Discogs*: *Blues*, *Brass & Military*, *Children's*, *Non-Music*, *Stage & Screen*).

# genres	boosting rounds [50, 2000]	tree depth [3, 15]	feature fraction [1, # of predictors]	leaf size [1, 100]	learning rate [0.001, 0.1]	target ratio [1, max/min]
5	93	12	68	46	0.072	4.815
12	1145	10	5	28	0.046	2.116
25	1512	9	85	91	0.005	2.641
32	176	9	55	15	0.0182	4.589

Table 2: Hyperparameter settings for final LightGBM models at different granularity levels.

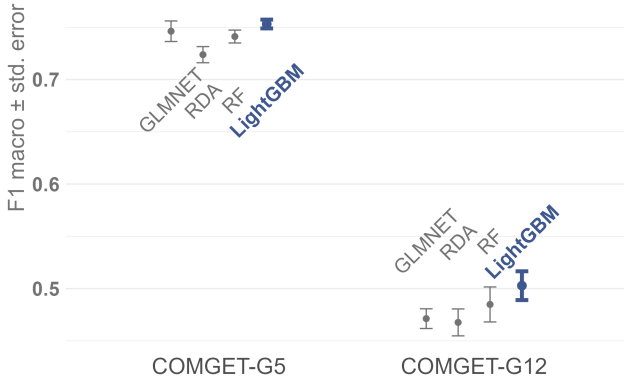


Figure 5: Performance comparison of different learners (GLMNET, RDA, RF, GBM) on 20% of the data set for low and medium genre granularity; plots show mean and standard error of $F1_{\text{macro}}$ across CV folds.

pop (21%) and *rock* (19%). The fact that electronic music’s stylistic elements nowadays frequently appear across pop and rock subgenres, explaining the observed misclassifications. Similarly, *metal* (61% TP) misclassifications with *rock* (37%) reflect their close relationship.

For the POPTRAG dataset classified with COMGET-G12 we achieved $F1_{\text{macro; CV}} = 0.519 \pm 0.007$ and $F1_{\text{macro; hold-out}} = 0.544$ ($\kappa_{\text{CV}} = 0.546 \pm 0.010$; $\kappa_{\text{hold-out}} = 0.551$), which parallels the lower *community certainty* (67%), indicating larger classification difficulty for both, listener community and algorithmic classifier. Also in this analysis, per-class F1 and *certainty* are highly correlated ($\text{Cor}_{\text{Kendall}} = 0.721$). The confusion matrix (Figure 6b) shows primary confusion among *rock subgenres* (*alternative rock*, *hard rock*, *indie rock*, *pop rock*). *Heavy metal* (56% TP) is frequently confused with *metal* (34%) but less often vice versa (*metal* 71% TP, 13% confusion with *metal*), and sometimes with *hard rock* (10% of *hard rock* misclassified as *heavy metal*), despite being recognized as distinct from *rock* (3% confusion). *indie rock* (49% TP) and *jazz* (61% TP) are often misclassified as *electronic* (*indie rock* 13%, *jazz* 11%); *jazz* is also confused with *pop* (14%). *pop rock* performs worst ($F1 = 0.182$), matching its low *community certainty* (50%). It is primarily confused with high-certainty genres *rock* (22%) and *pop* (34%), suggesting *pop rock* shares ambiguous characteristics with both. Conversely, *schlager* and *hip hop* tracks are classified quite well ($F1 = 0.837$ and 0.858 , respectively) despite differing *community certainties* (67% and 100%), indicating these genres possess very distinct characteristics in the feature space.

For the POPTRAG dataset classified with COMGET-G25 we obtained further loss in classification performance ($F1_{\text{macro; CV}} = 0.354 \pm 0.006$; $F1_{\text{macro; hold-out}} = 0.348$; $\kappa_{\text{CV}} = 0.436 \pm 0.009$; $\kappa_{\text{hold-out}} = 0.430$) alongside decreased *community certainty* (56%). The correla-

tion between *certainty* and F1 weakened ($\text{Cor}_{\text{Kendall}} = 0.589$), is still high, suggesting other factors might degrade classification performance at higher granularity. The confusion matrix (see appendix figure B.2a) again reveals persistent *rock subgenre* confusion (*alternative rock*, *blues rock*, *classic rock*, *folk rock*, *pop rock*) and *metal*-related confusion (*death metal*, *folk rock*, *hard rock*, *heavy metal*, *power metal*). A secondary confusion center emerges around *pop*, which frequently misclassifies with *alternative rock*, *classical*, *electronic*, *folk rock*, *pop rock*, *r&b*, *singer-songwriter*, *soul*, and *synth-pop*. The model performs well ($F1 > 0.6$) on *classical*, *electronic*, *hip hop*, *metal*—genres with distinct feature signatures and moderate to high *community certainty* (ranging 67%-100%). Poor performers ($F1 < 0.2$; *alternative rock*, *blues rock*, *classic rock*, *folk*, *folk rock*, *pop rock*, *power metal*, *certainty* ranging 33%-50%) represent fuzzy genres difficult to classify for both community and model.

For the POPTRAG dataset classified with COMGET-G32, where a strongly decreased community certainty could be observed (54%), the model for showed a respective lower classification performance ($F1_{\text{macro; CV}} = 0.344 \pm 0.003$; $F1_{\text{macro; hold-out}} = 0.332$; $\kappa_{\text{CV}} = 0.392 \pm 0.005$; $\kappa_{\text{hold-out}} = 0.339$, see appendix figure B.2b). However, correlation between *certainty* and F1 declined further ($\text{Cor}_{\text{Kendall}} = 0.560$), indicating that *community uncertainty* is not the only source of lower classification performance here. Given similarity to COMGET-G25, please refer to the appendix for a detailed confusion matrix.

Figure 7 shows the top 20 most important features of the developed COMGET-G5 and COMGET-G12 algorithmic classifiers (for COMGET-G25 and COMGET-G32 feature importances see appendix figure B.2). While we find similar features in the top 20 across all four levels of granularity (e.g., *acousticness* (*Echo-Nest*), *artist distributed as “metal”*, *age of artist*), the importance of single features differs. For COMGET-G5, features that describe music style characteristics (e.g., *acousticness*, *speechiness*, *explicitness of lyrics*) and artist persona (e.g., *single artist vs. band*, *age of artist*) are most important. For COMGET-G12, *popularity* features and the *release year* are additionally taken into account. For COMGET-G25 and COMGET-G32 the most important features are mostly artist-related, indicating that fine-grained genre distinctions rely strongly on release-milieu signals rather than music style represented by high-level audio features.

5. Discussion

With the present article, we introduce a novel and methodologically transparent procedure for constructing well-balanced hierarchical genre taxonomies and according genre category assignments from listener-community discourse on genres represented in folksonomies on open music databases. Specifically, we

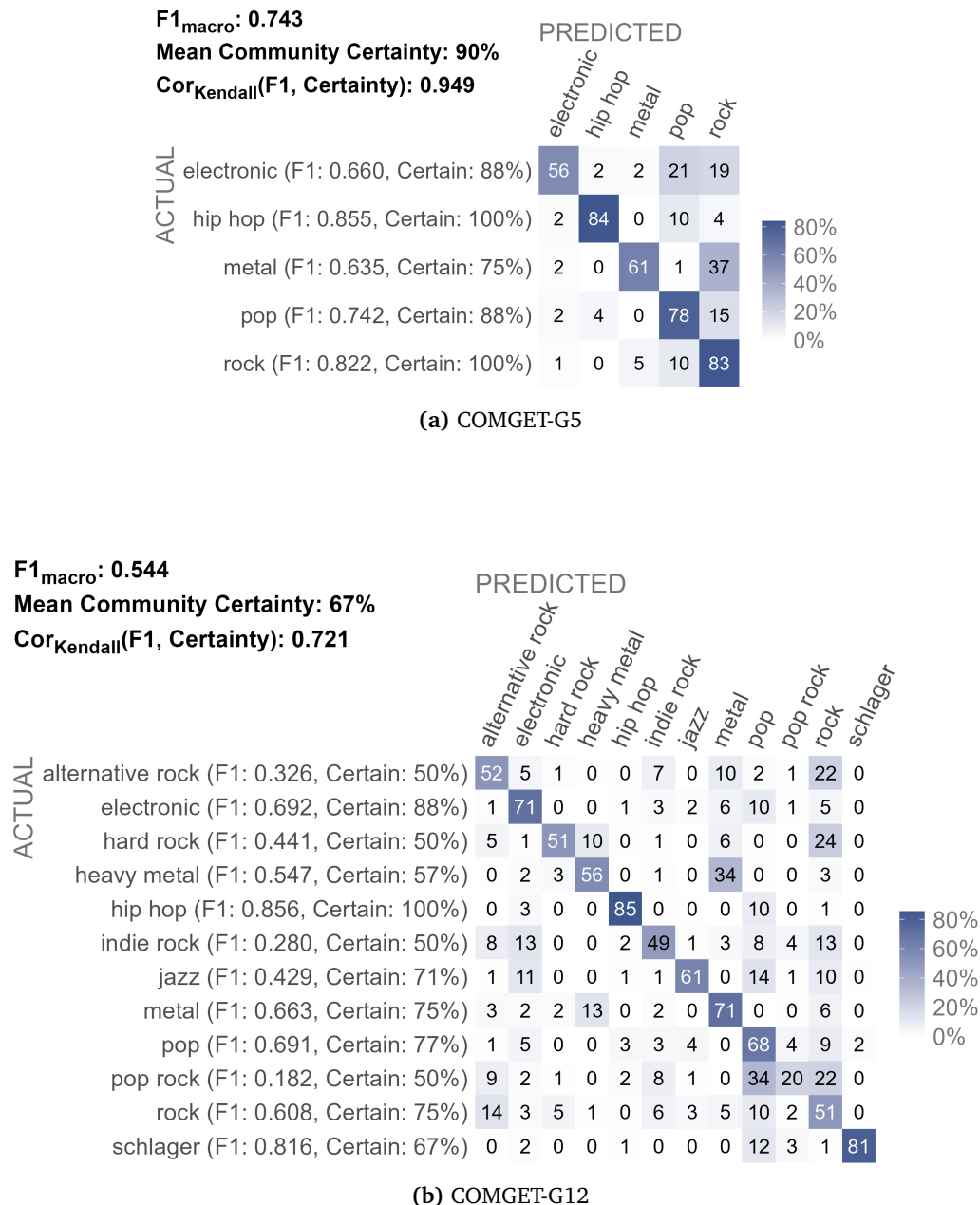


Figure 6: Confusion matrices for the final GBM models on the hold-out set. Values show $P(\text{Predicted}|\text{Actual})$; *certainty* indicates the mean assignment probability for each *genre category* across community votes.

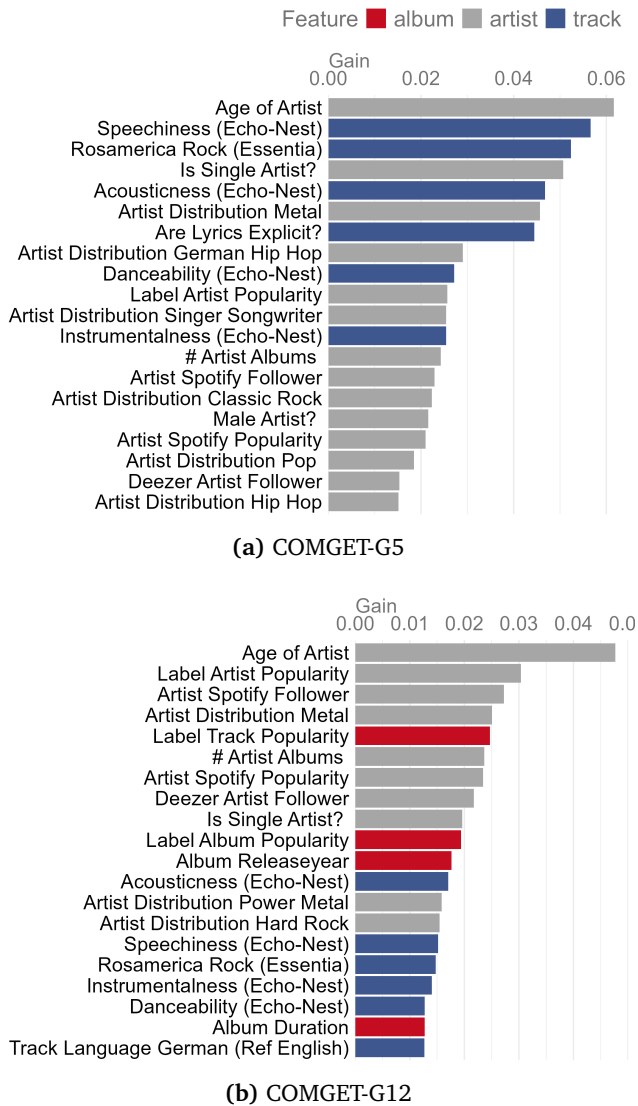


Figure 7: Top 20 most important features for automated classification at low and medium granularity. Values show relative feature importance from the final LightGBM models.

make three methodological key contributions: (RQ1) an algorithm to construct a balanced hierarchical genre tree (COMGET) from folksonomy genre tag co-occurrences, (RQ2) a folding algorithm (FOLDCAT) that generates optimally balanced hierarchical genre categories and final track-assignments at user-specified granularities from such trees. Finally, exemplarily employing a large German popular music corpus (POPTRAG) and MusicBrainz folksonomy votes, we (RQ3) demonstrate that resulting community-based genre assignments are a) comparable in content to existing and frequently-used genre taxonomies (i.e. *Discogs*, *Deezer*) while exhibiting more desirable features with a German popular music corpus in terms of category balance and category coverage and b) can be learned well with supervised machine learning by drawing on openly available music descriptors only. Hence, we recommend our novel approach as a transparent, reproducible, and empirically grounded alternative to commercial or expert-curated genre taxonomies in future applied or basic research. The method can in principle be adopted to any other music corpus and other genre tagging systems that provide co-occurrences of tags (e.g. *Spotify*, *YouTube*, *SoundCloud*). Also, while we focussed our exemplary analysis on track genre, the approach can also be used for assigning genres to artists or releases in a similar way.

5.1 Benefits of the presented approach for Computational Musicology and Cultural Analysis

Our COMGET analysis of the POPTRAG dataset of German popular music (RQ1) illustrates the method's capacity to computationally uncover popular music genre discourse structures. *Rock*, *pop*, and *hip hop* emerged as three primary clusters for German popular music genres in Germany, partially aligning with Silver et al. (2016) who identified three main international clusters analyzing MySpace artist networks in 2007. However, structures differ: *rock* forms a distinct cluster in POPTRAG versus Silver et al.'s scattered "Niche" genres; *hip hop* shows minimal differentiation (consistent with Silver et al.); *pop* exhibits stylistic diversity versus Silver et al.'s "Rock 'n Roll" category. Direct comparison remains limited—Silver analyzed artists on a social networking platform, we analyze tracks in a national listening corpus across different decades—yet the method enables such comparative analyses across corpora, revealing how genre discourse varies by platform, national context, and historical period (Pichl et al., 2017; Nie, 2022).

Crucially, hierarchical COMGET genre taxonomies as obtained in results of RQ1 also capture how contemporary music listeners in general categorize popular music rather than musicological historical genealogies (Pachet and Cazaly, 2000) do. While the COMGET-G reveals indeed some unambiguous genealogical links (*rock* → *alternative rock*), we also ob-

tained some at-first-glance counterintuitive links that rather invite a popular music discourse interpretation rather than genealogical correction from a musicological stance. For instance, *classical* and *jazz* emerge as *pop* subgenres in the COMGET-G tree. Two factors explain this placement: POPTRAG mostly contains very accessible, popular and partly even crossover tracks from these genres —popular opera arias, smooth jazz compilations, chart-oriented classical crossover artists. This corpus profile combined with listeners' tendency to apply broad labels when tagging popular music (Bogdanov et al., 2019) may explain the findings. Similarly, both *heavy metal* and *metal* appear as sibling *rock* subgenres, reflecting shared historical roots in 1970s psychedelic and hard rock while acknowledging *metal's* later distinctiveness (Gracyk, 2016).

HIER FEHLT NOCH KOMPLETT BENEFITS RQ2, zum Beispiel die Anwendung im DFG Projekt, dass nman kategorienzahl für Diversitätsanalysen von Playlists oder für Musikgeschmack beliebig skalieren kann, das man interessant Subgenre-Analysen machen kann, historische Entwicklung des Auftauchens und der Dominanz von Genres in den Chartts untersuchen kann und vieles mehr...

Finally, both, COMGET (RQ1) and FOLDCAT (RQ2) consequently implement power-critical concepts from popular music studies and cultural analysis (du Gay et al., 2013) into computational methods for making them applicable with large popular music corpora. Specifically, our approach employs radically democratic principles throughout: community votes determine co-occurrences, strongest links define hierarchies, majority voting assigns single labels. Resulting community genre taxonomies, genre categories, and genre category assignments reflect collective listener discourse rather than industry or musicological expert impositions, democratizing genre knowledge production for computational musicology. We envision that this will improve the validity of music-preference studies by stronger aligning measurements with how listeners actually categorize music.

Lastly, feature Importances obtained when developing algorithmic classifiers for the comget solutions of different granularities (RQ3) reveal how listener communities actually perceive popular music genres: artist-level features dominate predictions across all granularities tested, with distribution genres and artist demographics outweighing music style attributes —a finding consistent with popular music genre theory emphasizing release milieu (Neale, 1980). Music stylistic attributes play a secondary role: They only help to distinguish broad categories of music, then metadata and lyrics appear to be important for differentiation on lower hierarchical levels of genres. Popularity gains importance at finer resolutions as a genre signal itself (Holt, 2007). Disappointingly, our but

bag-of-words and lexical lyric features worked rather weak, pointing toward LLM-based embeddings as a promising direction (Prevedello et al., 2024). Genre-specific confusion patterns further illuminate boundary strength: e.g., *rock* subgenre overlap (*alternative*, *pop rock*, *indie*) reveals fuzzy boundaries; *electronic* permeability across *pop*, *jazz*, and *indie* reflects stylistic fluidity; and *schlager-hip hop* separation demonstrates distinct release milieus and stylistic features.

5.2 Benefits of the presented for MIR Research and Recommender Applications

A core methodological drawback in MIR is the fact that most hitherto genre recognition approaches lack explicit theoretical grounding (Green et al., 2024). Conversely, our approach presented here operationalizes genre-as-discourse (Foucault, 2013; Neale, 1980), treating proper genre assignment as an empirical question rather than an ontological given. Also we theoretically ground the choice of audio features and metadata and lyric features (XXX anonymous author paper). This connects computational methods with popular music studies and enables the MIR community to engage with humanities scholarship regarding the broader topic of cultural categorization (DiMaggio, 1987).

Furthermore, our single-folksonomy approach ensures epistemic consistency. Existing mixed-source methods conflate different communities' genre concepts across platforms with varying demographics, tag voting systems, tag pools, and discursive contexts, introducing uncontrolled variance (Schreiber, 2015; Bogdanov et al., 2019). Conversely, single-source taxonomies eliminate cross-platform variance, ensuring all genre assignments represent a specific community's discourse rather than mixing incompatible categorization systems, thereby addressing issues of representativeness in MIR (XXX). By refraining from introducing any arbitrary numeric thresholds throughout the genre categorization pipeline and drawing on open-source implementation, our approach can be simply reproduced and adapted to any music corpus and multi-genre-tag system, as we have demonstrated by using it for constructing industry distribution genres from Spotify tags in RQ3 of this paper. Researchers may replicate the method with their target population's folksonomy, generating taxonomies specific to their chosen listener community.

FOLDCAT addresses diverse practical requirements for handling complex hierarchical genre taxonomies without any unnecessary information loss while enabling cross-study comparison. The folding algorithm generates optimally balanced genre categories at multiple granularities using a single statistical criterion (Gini coefficient) across hierarchy levels. In addition, underlying genre category assignments stay probabilistic. Hence, the framework quantifies achievable per-

formance ceilings through community certainty, since models cannot exceed human consistency (Sturm and Flexer, 2023). Obtained large correlations between certainty and classifier performance validate this approach, grounding MIR in listener perception limits. Obtained classifier performance demonstrates general learnability of the identified genre categories, while systematic confusion patterns reveal fuzzy boundaries (*rock* subgenres), stylistic permeability (*electronic* across *pop/jazz/indie*), and hierarchical nuance (*heavy metal* versus *metal* asymmetry). These patterns inform which granularities may suit different research questions and reaffirm that genre labels remain learnable in principle by drawing on a theoretically proposed set of features and metadata suggested by the anonymized authors.

5.3 Limitations

Several constraints limit interpretation and generalization of our exemplary results. For our exemplary analysis with POPTRAG dataset we drew on MusicBrainz community votes. This crowdbased music database was launched in 2000 and does not provide individual vote timestamps, so COMGET-G aggregates roughly 25 years of listener votes spanning a long time period across which the perception of genre boundaries might have most likely shifted gradually. Furthermore, platform user demographics and expertise remain opaque (approximately 1.2 million anonymous editor accounts; MusicBrainz, nd), preventing assessment of a possibly sociodemographic or expertise bias. For the present analysis we had to exclude 36% tracks of our POPTRAG dataset due to insufficient genre tags or low number of votes, thereby introducing a substantial popularity bias to our analysis—which is probable acceptable for our popular music focus but possibly problematic for analyses directed at more niche genres. Radical-democratic reduction of multi-tag may have reduced nuances, though this simplification proves necessary for many applied research tasks and aligns with our theory. Methodologically, collinearity among predictors complicates feature importance interpretation.

5.4 Future Directions

Future work will pursue into three different directions. First, we plan to convene German popular music experts to review obtained COMGET-G taxonomy. Thereto, we will label prototypical tracks according to our machine learning with genre categories at different levels of granularity, and identify misassignments to strengthen COMGET-G's validity. Second, we plan to replicate the pipeline on independent folksonomies and regional corpora of tracks (e.g., US and UK charts) to reveal how national listening cultures reshape genre structure. Third, we are aiming at enhancing classification performance further by employing LLMs to produce advanced lyric features.

Beyond MIR and computational musicology, the approach presented may also offer new possibilities benefits for stakeholders across the music industry: radio programmers, playlist editors, and music librarians to gain insight into how audiences organize music; A&R teams to identify gaps between industry positioning and listener reception; marketing departments to recognize which subcultural boundaries audiences perceive versus industry-imposed categories; music journalists to ground critical language in empirical listener consensus; and platform designers to align recommendation logic with listeners' mental models.

Competing Interests

The authors have no competing interests to declare.

References

- Airolidi, M., Beraldo, D., and Gandini, A. (2016). Follow the algorithm: An exploratory investigation of music on YouTube. *Poetics*, 57:1–13. <https://doi.org/10.1016/j.poetic.2016.05.001>.
- Alonso-Jiménez, P., Bogdanov, D., Pons, J., and Serra, X. (2020). Tensorflow audio models in essentia. In *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 266–270.
- Bogdanov, D., Porter, A., Schreiber, H., Urbano, J., and Oramas, S. (2019). The Acousticbrainz Genre Dataset: Multi-Source Multi-Level, Multi-Label, and Large-Scale. In *20th International Society for Music Information Retrieval Conference*, Delft.
- Bogdanov, D., Wack, N., Gómez, E., Gulati, S., Herrera, P., Mayor, O., Roma, G., Salamon, J., Zapata, J., and Serra, X. (2013). ESSENTIA: an open-source library for sound and music analysis. In *Proceedings of the 21st ACM international conference on Multimedia*, pages 855–858, New York. Association for Computing Machinery.
- Bonini, T. and Magaudda, P. (2024). *Platformed! How Streaming, Algorithms and Artificial Intelligence are Shaping Music Cultures*. Springer Nature Switzerland. <https://10.1007/978-3-031-43965-0>.
- Bourdieu, P. (1985). The market of symbolic goods. *Poetics*, 14(1):13–44.
- Brackett, D. (2016). *Categorizing Sound: Genre and Twentieth-Century Popular Music*. University of California Press. <https://doi.org/10.1525/california/9780520248717.001.0001>.
- Brand, C. O., Acerbi, A., and Mesoudi, A. (2019). Cultural evolution of emotional expression in 50 years of song lyrics. *Evolutionary Human Sciences*, 1:e11. <https://doi.org/10.1017/ehs.2019.11>.
- Caparrini, A., Arroyo, J., Pérez-Molina, L., and Sánchez-Hernández, J. (2020). Automatic sub-genre classification in an electronic dance music taxonomy. *Journal of New Music Re-*

- search, 49(3):269–284. <https://doi.org/10.1080/09298215.2020.1761399>.
- Carbone, L., Alvarez-Cueva, P., and Vandenbosch, L. (2024). Status Markers in Popular Music Across Six Countries: A Content Analysis of Gender, Race/Ethnicity, Genre, and Capital in Music Lyrics. *Sex Roles*, 90(7):891–909. <https://doi.org/10.1007/s11199-024-01483-0>.
- Crauwels, K. (2022). Musicmap. the genealogy and hierarchy of musical genres. <https://musicmap.info/>.
- Deezer (n.d.). Deezer music streaming platform. Retrieved 2025-12-12 from <https://developers.deezer.com/login?redirect=/api>.
- DiMaggio, P. (1987). Classification in Art. *American Sociological Review*, 52(4):440. <https://doi.org/10.2307/2095290>.
- Discogs (2024). Database guidelines. 9. genres and styles. <https://support.discogs.com/hc/en-us/articles/360005055213-Database-Guidelines-9-Genres-Styles>.
- du Gay, P., Hall, S., Janes, L., Madsen, A. K., Mackay, H., and Negus, K. (2013). *Doing Cultural Studies: The Story of the Sony Walkman*. SAGE.
- Fabbri, F. (1982). A Theory of Musical Genres: Two Applications. In Horn, D. and Tagg, P., editors, *Popular Music Perspectives*, pages 52–81. IASPM.
- Flexer, A., Dörfler, M., Schlüter, J., and Grill, T. (2018). Hubness as a Case of Technical Algorithmic Bias in Music Recommendation. In *2018 IEEE International Conference on Data Mining Workshops (ICDMW)*. <https://doi.org/10.1109/ICDMW.2018.00154>.
- Flexer, A. and Grill, T. (2016). The Problem of Limited Inter-rater Agreement in Modelling Music Similarity. *Journal of New Music Research*, 45(3):239–251. <https://doi.org/10.1080/09298215.2016.1200631>.
- Flexer, A. and Schnitzer, D. (2010). Effects of Album and Artist Filters in Audio Similarity Computed for Very Large Music Databases. *Computer Music Journal*, 34(3):20–28. <https://www.jstor.org/stable/40963029>.
- Foucault, M. (2013). *Archaeology of Knowledge*. Routledge, London, 2 edition. <https://doi.org/10.4324/9780203604168>.
- Frith, S. (1996). *Performing Rites: On the Value of Popular Music*. Harvard University Press.
- Genius (2009). About genius. <https://genius.com/Genius-about-genius-annotated>.
- Genussov, M. and Cohen, I. (2010). Musical genre classification of audio signals using geometric methods. In *18th European Signal Processing Conference*, pages 497–501. <https://ieeexplore.ieee.org/document/7096332>.
- Gracyk, T. (2016). Heavy metal: Genre? Style? Subculture? *Philosophy Compass*, 11(12):775–785. <https://doi.org/10.1111/phc3.12386>.
- Green, O., Sturm, B. L., Born, G., and Wald-Fuhrmann, M. (2024). A critical survey of research in music genre recognition. In *25th International Society for Music Information Retrieval Conference*, San Francisco. https://ismir2024program.ismir.net/poster_149.html.
- Han, H., Wang, W.-Y., and Mao, B.-H. (2005). Borderline-SMOTE: A New Over-Sampling Method in Imbalanced Data Sets Learning. In Huang, D.-S., Zhang, X.-P., and Huang, G.-B., editors, *Advances in Intelligent Computing*, pages 878–887, Berlin, Heidelberg. Springer. https://doi.org/10.1007/11538059_91.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009). Model Assessment and Selection. In Hastie, T., Tibshirani, R., and Friedman, J., editors, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, pages 219–259. Springer, New York. https://doi.org/10.1007/978-0-387-84858-7_7.
- Hildebrandt, A., Jäckle, S., Wolf, F., and Heindl, A. (2015). Inhaltsanalyse. In *Methodologie, Methoden, Forschungsdesign*, pages 299–333. Springer Fachmedien Wiesbaden, Wiesbaden. https://doi.org/10.1007/978-3-531-18993-2_12.
- Holt, F. (2007). Genre in Popular Music. In *Genre in Popular Music*. University of Chicago Press. <https://doi.org/10.7208/9780226350400>.
- Husslage, B. G. M., Rennen, G., van Dam, E. R., and den Hertog, D. (2011). Space-filling Latin hypercube designs for computer experiments. *Optimization and Engineering*, 12(4):611–630. <https://doi.org/10.1007/s11081-010-9129-8>.
- Japkowicz, N. and Stephen, S. (2002). The class imbalance problem: A systematic study. *Intelligent Data Analysis*, 6(5):429–449.
- Jiang, J., Ponnada, A., Li, A., Lacker, B., and Way, S. F. (2024). A Genre-Based Analysis of New Music Streaming at Scale. In *Proceedings of the 16th ACM Web Science Conference*, pages 191–201, New York. Association for Computing Machinery. <https://dl.acm.org/doi/10.1145/3614419.3644002>.
- Lange, E. B., Gernandt, E., and Merrill, J. (2025). Categorizing music by genres. *Annals of the New York Academy of Sciences*, page nyas.15404. <https://doi.org/10.1111/nyas.15404>.
- Lena, J. C. and Peterson, R. A. (2008). Classification as Culture: Types and Trajectories of Music Genres. *American Sociological Review*, 73(5):697–718. <https://doi.org/10.1177/000312240807300501>.
- Lonsdale, A. J. (2021). Musical taste, in-group favoritism, and social identity theory: Re-testing the predictions of the self-esteem hypothesis. *Psychology of Music*, 49(4):817–827. <https://doi.org/10.1177/0305735619899158>.

- McDonald, G. (n.d.). Every noise at once. Retrieved 2024-04-23 from <https://everynoise.com/>.
- Merlini, M. (2020). A critical (and interdisciplinary) survey of popular music genre theories. *Interdisciplinary Studies in Musicology*, 20:79–94. <https://www.ceeol.com/search/article-detail?id=1105289>.
- MusicBrainz (n.d.). Database statistics. Retrieved 2025-12-12 from <https://musicbrainz.org/statistics/>.
- Neale, S. (1980). *Genre*. British Film Institute.
- Negus, K. (1999). *Music Genres and Corporate Cultures*. Routledge, London. <https://doi.org/10.4324/9780203169469>.
- Nie, K. (2022). Inaccurate Prediction or Genre Evolution? Rethinking Genre Classification. *Zenodo*. <https://doi.org/10.5281/zenodo.7316661>.
- Pachet, F. and Cazaly, D. (2000). A taxonomy of musical genres. In *Content-Based Multimedia Information Access - Volume 2*, RIAO '00, pages 1238–1245, Paris, FRA. <https://dl.acm.org/doi/10.5555/2856151.2856177>.
- Pichl, M., Zangerle, E., Specht, G., and Schedl, M. (2017). Mining Culture-Specific Music Listening Behavior from Social Media Data. In *IEEE International Symposium on Multimedia (ISM)*, pages 208–215.
- Porcaro, L., Gómez, E., and Castillo, C. (2024). Assessing the Impact of Music Recommendation Diversity on Listeners: A Longitudinal Study. *ACM Trans. Recomm. Syst.*, 2(1):3:1–3:47. <https://doi.org/10.1145/3608487>.
- Prevedello, G., Blin, I., Monechi, B., and Ubaldi, E. (2024). Lyrics for Success: Embedding Features for Song Popularity Prediction. In Kruspe, A., Oramas, S., Epure, E. V., Sordo, M., Weck, B., Doh, S., Won, M., Manco, I., and Meseguer-Brocal, G., editors, *Proceedings of the 3rd Workshop on NLP for Music and Audio (NLP4MusA)*, pages 75–80, Oakland. Association for Computational Linguistics. <https://aclanthology.org/2024.nlp4musa-1.13/>.
- RateYourMusic (n.d.). All genres. Retrieved 2025-12-12 from <https://rateyourmusic.com/genres/>.
- Rentfrow, P. J., Goldberg, L. R., and Levitin, D. J. (2011). The structure of musical preferences: A five-factor model. *Journal of Personality and Social Psychology*, 100(6):1139–1157. <https://doi.org/10.1037/a0022406>.
- Schreiber, H. (2015). Improving Genre Annotations for the Million Song Dataset. In *16th International Society for Music Information Retrieval Conference*.
- Schreiber, H. (2016). Genre Ontology Learning: Comparing Curated with Crowd-Sourced Ontologies. In *17th International Society for Music Information Retrieval Conference*, New York.
- Shi, Y., Ke, G., Soukhavong, D., Lamb, J., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y., Titov, N., and Cortes, D. (2025). *lightgbm: Light Gradient Boosting Machine*. R package version 4.6.0.99.
- Silver, D., Lee, M., and Childress, C. C. (2016). Genre Complexes in Popular Music. *PLOS ONE*, 11(5):e0155471. <https://doi.org/10.1371/journal.pone.0155471>.
- Snoek, J., Larochelle, H., and Adams, R. P. (2012). Practical Bayesian Optimization of Machine Learning Algorithms. In *Advances in Neural Information Processing Systems*, volume 25. Curran Associates, Inc.
- Sordo, M. (2008). The quest for musical genres: Do the experts and the wisdom of crowds agree? In *9th International Society for Music Information Retrieval Conference*, Philadelphia.
- Sturm, B. L. T. and Flexer, A. (2023). A Review of Validity and Its Relationship to Music Information Research. In *24th International Society for Music Information Retrieval Conference*, Milan. <https://doi.org/10.5281/ZENODO.10265219>.
- Tay, J. K., Narasimhan, B., and Hastie, T. (2023). Elastic net regularization paths for all generalized linear models. *Journal of Statistical Software*, 106(1):1–31.
- Tzanetakis, G. and Cook, P. (2002). Musical genre classification of audio signals. *IEEE Transactions on Speech and Audio Processing*, 10(5):293–302. <https://doi.org/10.1109/TSA.2002.800560>.
- Valverde, R. C. (2025). Inequity by design: Music streaming taxonomies as ruinous infrastructure. In Hesmondhalgh, D., editor, *Music Streaming around the World*, pages 191–210. University of California Press, 1 edition. <https://doi.org/10.2307/jj.33903674.16>.
- van Venrooij, A. and Schmutz, V. (2018). Categorical ambiguity in cultural fields: The effects of genre fuzziness in popular music. *Poetics*, 66:1–18. <https://doi.org/10.1016/j.poetic.2018.02.001>.
- Weihs, C., Ligges, U., Luebke, K., and Raabe, N. (2005). klar analyzing german business cycles. In Baier, D., Decker, R., and Schmidt-Thieme, L., editors, *Data Analysis and Decision Support*, pages 335–343, Berlin. Springer-Verlag.
- Wright, M. N. and Ziegler, A. (2017). ranger: A fast implementation of random forests for high dimensional data in C++ and R. *Journal of Statistical Software*, 77(1):1–17.
- Zhang, J. and Mani, I. (2003). kNN Approach to Unbalanced Data Distributions: A Case Study involving Information Extraction. In *Proceedings of workshop on learning from imbalanced datasets*, pages 1–7, Ottawa. ICML. <https://www.site.uottawa.ca/~nat/Workshop2003/jzhang.pdf>.